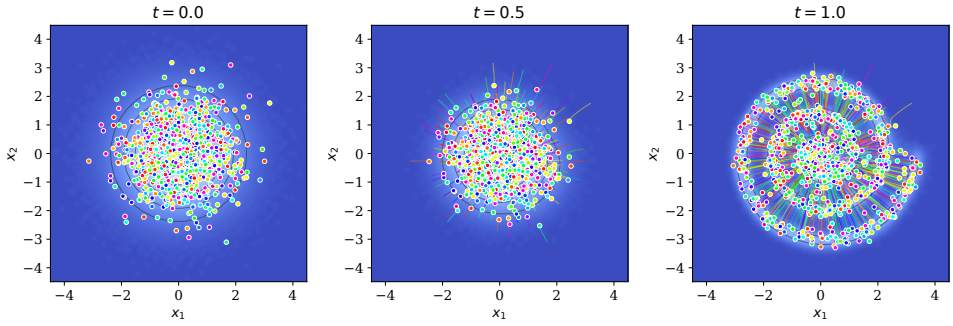
WHEN GENERATIVE MODELS DO NOT EXHIBIT DOUBLE DESCENT: A FLOW MATCHING EXAMPLE

Pierre Guilbert

Abstract.

1. Introduction

Generative modelling is a field of machine learning that aims at approximating an unknown data distribution, with density p_{data}, from a finite set of independent samples $(x_1, \dots, x_N) \in E^N$, where $E = \mathbb{R}^d$. Methods include generative adversarial networks (GANs) [29] and variational autoencoders [30]. Diffusion models [20, 21] and Flow Matching [2] tackle modelling and sampling together by learning an evolution from a simple initial distribution to the data distribution. We are interested here in Flow Matching.



Open the particle video (MP4)

Figure 1. Gaussian-to-spiral transport learnt with fresh training samples. The video follows 975 particles; click the preview to open the accompanying MP4. The example is described in Section 3.

Let p_{init} be the initial density, and let $Z \sim p_{\text{init}}$ and $X \sim p_{\text{data}}$ be independent. We consider the path of random variables

$$X_t = tX + (1-t)Z, \qquad 0 \le t \le 1. \tag{1}$$

For $0 < t < 1$, its density p_t is the convolution of the data and initial densities contracted by t and $1-t$, respectively [28]:

$$p_t = \left[\frac{1}{t^d} p_{\text{data}}\left(\frac{\cdot}{t}\right)\right] * \left[\frac{1}{(1-t)^d} p_{\text{init}}\left(\frac{\cdot}{1-t}\right)\right]. \tag{2}$$

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Flow Matching consists in learning a velocity field $v_\theta : [0,1] \times E \to E$ to transport p_{init} to p_{data}. The target velocity v_t and the density path are linked by the continuity equation [2, 28]:

$$\partial_t p_t = -\operatorname{div}(p_t v_t). \tag{3}$$

Sampling from the learnt distribution is then performed using a classical numerical integration solver, such as Euler or Runge–Kutta, applied to the learnt velocity vector field with initial samples drawn from p_{init}.

2. Double descent and memorisation

In supervised machine learning, it has been shown that, under some conditions, increasing model capacity can lead to a *double-descent* learning curve [31, 32]. The test error first decreases, increases near the interpolation threshold, where the model has enough capacity to fit the training samples, and decreases again for larger models. Nakkiran et al. [32] also observe double descent with longer training of a fixed ResNet18 on CIFAR-10 with label noise. Figure 2 illustrates both cases.

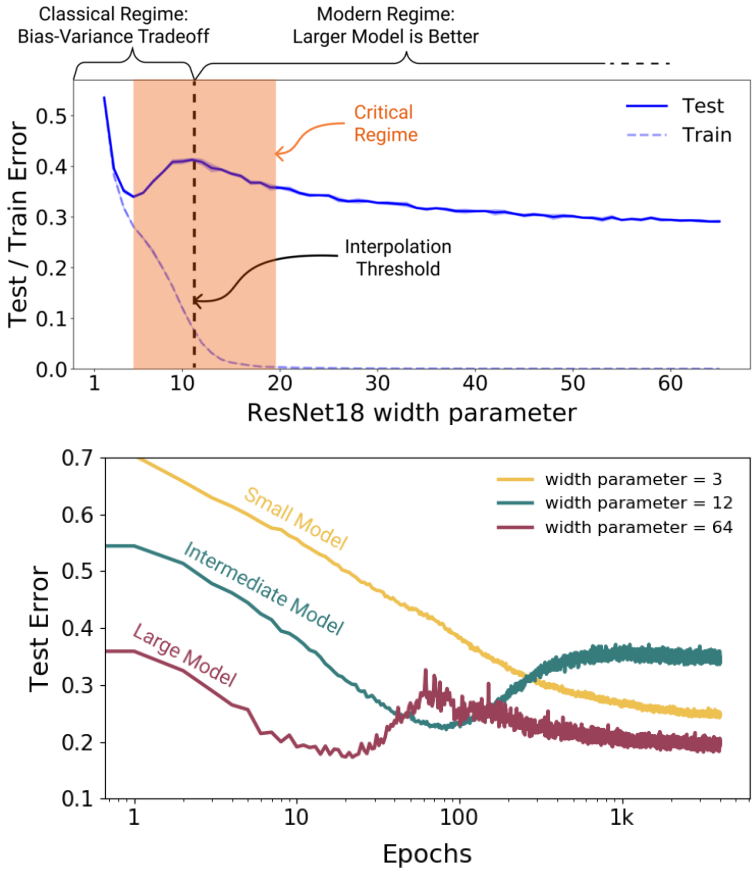


FIGURE 2. Double descent for ResNet18 models on CIFAR-10 with label noise. Top: increasing model width. Bottom: increasing training time for fixed architectures. Panels reproduced from Figures 1 (left) and 9 (left) of Nakkiran et al. [32].

Recent research on diffusion models reports a different behaviour. Farghly et al. [25] derive, in their analysis, a U-shaped test loss as a function of model capacity. Urfin et al. [26] find that increasing capacity can drive unregularised models towards memorisation. Along the training-time axis, Bonnaire et al. [24] identify an initial generalisation regime followed by a memorisation regime when using finite training data.

For Flow Matching trained on a fixed dataset, sampling X from the available points makes the prescribed endpoint the empirical measure

$$\widehat{P}_N = \frac{1}{N}\sum_{i=1}^{N} \delta_{x_i}. \tag{4}$$

Learning this empirical distribution more precisely concentrates generated samples around individual training points [2, 27]. Carré du champ Flow Matching proposes a way to regularise this quality–generalisation tradeoff by introducing anisotropic noise adapted to the local data geometry [27].

We will visualise the transition from generalisation to memorisation on a small two-dimensional example.

3. A two-dimensional visualisation

We propose to visualise this memorisation regime through a two-dimensional example. We construct p_{data} by smoothing the spiral

$$\gamma(s) = hs(\cos s, \sin s), \qquad 0 \le s \le 6\pi, \qquad h = \frac{1.1}{2\pi}, \tag{5}$$

with a Gaussian kernel of standard deviation 0.16. The curve parameter is sampled uniformly. The resulting density is restricted to $[-3.94, 3.94]^2$ and normalised. Figure 3 shows this density and a few samples.

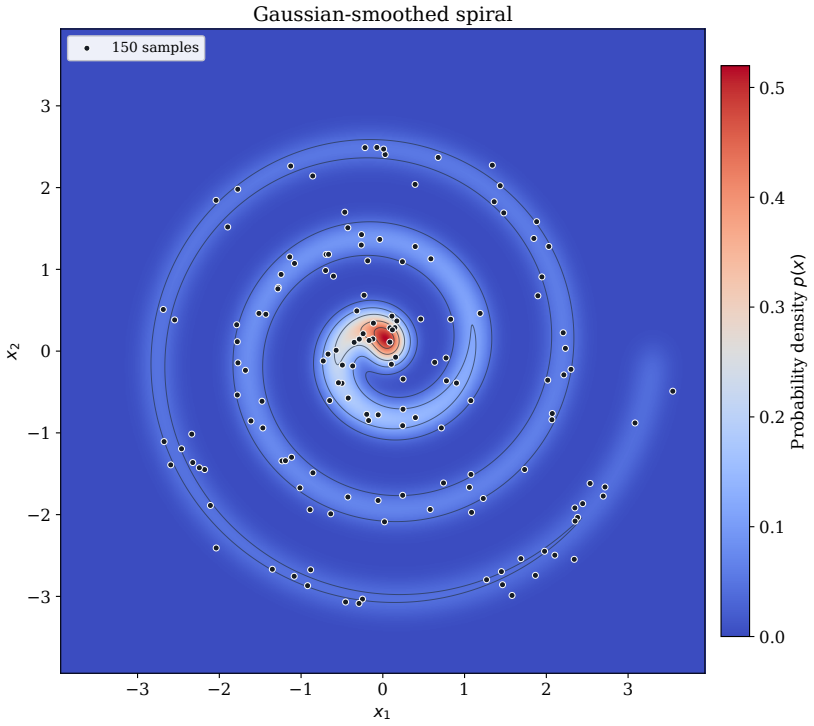


FIGURE 3. The spiral density p_{data} and 150 independent samples. Colours and contours represent the probability density.

3.1. Training from an infinite dataset

We first train an MLP with four hidden layers of width 128, SiLU activations and a sinusoidal time encoding. We use $p_{\mathrm{init}} = \mathcal{N}(0, \mathrm{Id})$. At each of the 20,000 training updates, we draw 1,024 fresh samples from p_{data}, independent Gaussian samples from p_{init}, and times uniformly in $[0, 1]$. Training therefore uses an on-demand stream of samples from the underlying distribution. The video in Figure 1 shows 975 initial Gaussian particles transported by the learnt field, using 50 fourth-order Runge–Kutta steps.

3.2. **Training from a finite dataset**

We now draw 1,000 independent samples from the same p_{data}: 500 are used for training and 500 for validation. These samples remain fixed for the full training. The finite-data model has four hidden layers of width 256, SiLU activations, the same time encoding, and 64 fixed spatial Fourier frequencies. Its architecture remains unchanged throughout this experiment.

We train the network for 100,000 epochs, where one epoch is a full pass over the 500 training samples, in two batches of 250. Gaussian samples and uniform times are drawn again at each visit.[1] Figure 4 shows the training and validation velocity-regression losses. For evaluation, we use 32 fixed noise/time draws per data point at every checkpoint.

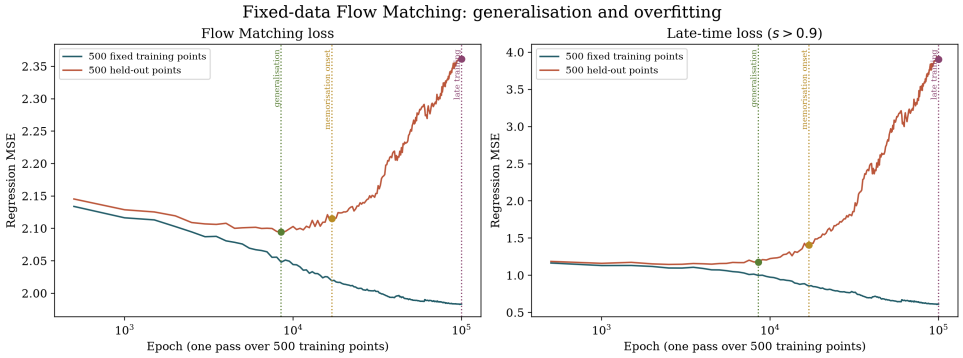
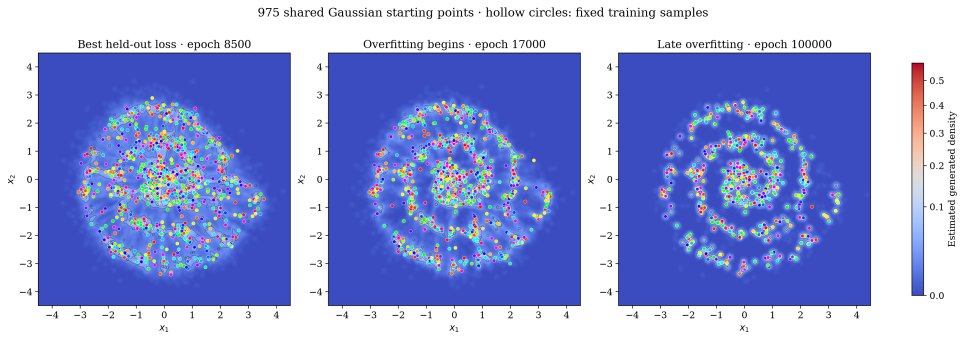


FIGURE 4. Training and validation losses for the fixed 500/500 split. Left: uniform times in $[0, 1]$. Right: times above 0.9. Vertical marks identify the three selected checkpoints.

The transported initial law at different training regimes is shown in Figure 5: generalisation at epoch 8,500, the beginning of overfitting at epoch 17,000, and memorisation at epoch 100,000. The three videos follow the same 975 particles sampled from p_{init}. As training continues, their final positions concentrate around the fixed training points.

[1] After epoch 30,000, updates use the exact empirical conditional-mean velocity to reduce gradient noise. This has the same Flow Matching optimum; the plotted losses keep the original conditional regression definition.



Generalisation Overfitting begins Memorisation

FIGURE 5. Transported densities at $t = 1$ for the three training regimes. Hollow circles mark the fixed training points. Click the labels to open the corresponding videos, or the image for the last video.

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